Law of sines

1

a False **b** False **c** True **d** True

2 No, because the law of sines requires one angle to find another angle.

3 Yes, because we can find the opposite angle to c using the angle sum of a triangle. Then we can find the other two sides using law of sines.

Unknown sides

4 $v = 10$

5

a $a = 10.5$ **b** $a = 6.9$ **c** $x = 13.1$ **d** $y = 4.4$

6

a $\angle A = 75.78^\circ$ **b** $b = 3.152\ \text{m}$ **c** $c = 5.723\ \text{m}$

7

a 29° **b** $c = 37.11$

8

a

i $13 < a < 47$ **ii** $a = 17.97$

b

i $17 < a < 63.2$ **ii** $a = 27.64$

9

a 39° **b** $k = 21.76$

Unknown angles

10

a $x = 30.4^\circ$ **b** $x = 32.7^\circ$ **c** $x = 26.0^\circ$ **d** $x = 43.2^\circ$

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11

a $x = 133.13^\circ$ **b** $x = 149.12^\circ$

Ambiguous case

12

a 0 **b** 1 **c** 1 **d** 2

13

a No **b** No **c** Yes **d** No

e Yes **f** Yes **g** No **h** Yes

14

a No **b** No **c** Yes **d** No

15

a $x = 41$ **b** 139°

Applications

16

a $a = 74^\circ$ **b** $d = 28.35$ m **c** $h = 16.66$ m

17

a $\angle ABC = 21.89^\circ$ **b** $\angle BCA = 123.11^\circ$

c $AB = 29.2$ km

18

a $\angle ADB = 23^\circ$ **b** $a = 35.65$ yd **c** $h = 20.4$ yd

19

a $\angle ACB = 102^\circ$ **b** $b = 226.55$ m **c** $h = 117$ m

20

a $\angle A = 38^\circ$ **b** $c = 22.71$ m **c** $b = 22.71$ m **d** 1.9 minutes

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